AWS EC2 Passwordless SSH and File Transfer Lab

Objective

The objective of this lab was to:

- Launch two Amazon EC2 instances using AWS.
- Establish passwordless SSH communication between the instances.
- Transfer files securely using SCP (Secure Copy Protocol).
- Understand SSH key-based authentication.

Prerequisites

- AWS Account

Step 1: Launch EC2 Instances

1. Open **AWS Management Console**.
2. Navigate to **EC2 Dashboard**.
3. Click **Launch Instance**.
4. Configure:
   - Number of Instances: **2**
   - AMI: **Amazon Linux**
   - Instance Type: **t2.micro or t3.micro**
5. Create a new Key Pair:
   - Name: `ec2-lab-key`
   - Type: RSA
   - Format: `.pem`
6. Download the generated key pair file:
   - `ec2-lab-key.pem`

Step 2: Configure Security Group

Create a Security Group with the following inbound rule:

Type	Protocol	Port	Source
SSH	TCP	22	0.0.0.0/0

Initially, SSH access was restricted to a single IP address, which prevented EC2 Instance Connect from working properly. Changing the source to `0.0.0.0/0` resolved the issue.

Step 3: Connect to EC2 Instances

Using **EC2 Instance Connect**:

1. Select the instance.
2. Click **Connect**.
3. Choose **EC2 Instance Connect**.
4. Username:

ec2-user

5. Click **Connect**.

Open separate browser tabs for both instances.

Step 4: Verify Private IP Addresses

On both instances:

hostname -I

Example:

Server 1: 172.31.22.108
Server 2: 172.31.xx.xxx

These private IPs are used for internal communication.

Step 5: Generate SSH Key Pair

On **Server 2**:

ssh-keygen -t rsa

Press Enter for all prompts:

Enter file in which to save the key:
Enter passphrase:
Enter same passphrase again:

Generated files:

~/.ssh/id_rsa
~/.ssh/id_rsa.pub

Step 6: View Public Key

On Server 2:

cat ~/.ssh/id_rsa.pub

Copy the complete output beginning with:

ssh-rsa

and ending with:

ec2-user@hostname

Step 7: Configure Passwordless Authentication

On **Server 1**:

Create or open the authorized keys file:

mkdir -p ~/.ssh
chmod 700 ~/.ssh

nano ~/.ssh/authorized_keys

Paste the copied public key into the file.

Save and exit:

Ctrl + O
Enter
Ctrl + X

Set permissions:

chmod 600 ~/.ssh/authorized_keys

Step 8: Test Passwordless SSH

From **Server 2**:

ssh ec2-user@<Server1_Private_IP>

Example:

ssh ec2-user@172.31.22.108

First-time connection:

Are you sure you want to continue connecting (yes/no)?

Type:

yes

Successful login without password confirms passwordless SSH setup.

Step 9: Test File Transfer Between Instances

Create a test file on Server 2:

echo "Hello AWS" > test.txt

Transfer the file to Server 1:

scp test.txt ec2-user@172.31.22.108:/home/ec2-user/

Verify on Server 1:

ls

Output:

test.txt

Passwordless file transfer is successfully established.

Step 10: Transfer Files from Local Machine to EC2

Using PowerShell:

scp -i ec2-lab-key.pem hello.txt ec2-user@51.20.80.47:/home/ec2-user/

Example:

scp -i ec2-lab-key.pem C:\Users\Harsh\Desktop\hello.txt
ec2-user@51.20.80.47:/home/ec2-user/

Successful output:

hello.txt 100%

Understanding the Keys Used

AWS Key Pair

ec2-lab-key.pem

Purpose:

- Authentication from Local Machine → EC2 Instance

Example:

ssh -i ec2-lab-key.pem ec2-user@51.20.80.47

SSH Key Pair

Generated using:

ssh-keygen -t rsa

Files:

id_rsa
id_rsa.pub

Purpose:

- Authentication from Server → Server

Example:

ssh ec2-user@172.31.22.108

without requiring a password.

Commands Used

hostname -I
ssh-keygen -t rsa
cat ~/.ssh/id_rsa.pub
nano ~/.ssh/authorized_keys
chmod 700 ~/.ssh

chmod 600 ~/.ssh/authorized_keys
ssh ec2-user@<private-ip>
scp test.txt ec2-user@<private-ip>:/home/ec2-user/

Learning Outcomes

After completing this lab, the following concepts were understood:

- AWS EC2 Instance Creation
- Security Groups and SSH Access
- Public and Private IP Addresses
- SSH Key-Based Authentication
- Passwordless SSH Communication
- SCP File Transfer
- Difference between AWS Key Pair (`.pem`) and SSH Key Pair (`id_rsa`, `id_rsa.pub`)
- Secure communication between cloud servers using SSH Protocol

Result

Successfully launched two Amazon EC2 instances, configured passwordless SSH authentication using RSA key pairs, and transferred files securely between instances using SCP without requiring passwords.